

1 Main Reference and Supplementary Resources

1.1 Main reference

1.1.1 Where can the main reference for the evaluation experiment be found?

4. No report describing the experiment is available and no pre-registration is not planned.

1.1.2 Which experiment is this form being completed for?

The reliability evaluation of the Hallucination Detector in Section 4.3 of the paper.

1.2 Supplementary resources

1.2 Where can the resources that were used in the evaluation experiment be found?

2. No resources shared.

1.3 Contact details

1.3.1 Details of the person completing this sheet.

1.3.1.1 Name of the person completing this sheet.

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1.3.1.2 Affiliation of the person completing this sheet.

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2 Evaluated System(s)

Questions 2.1–2.5 record information about the system(s) evaluated in the experiment that this sheet is being completed for. The input, output, and task questions in this section are closely interrelated: the value for one partially determines the others, as indicated for some combinations in Question 2.3.

2.1 What type of input do the evaluated system(s) take?

4. Text: subsentential unit of text
5. Text: sentence
6. Text: multiple sentences

2.2 What type of output do the evaluated system(s) generate?

5. Text: sentence.

2.3 What is the task that the evaluated system(s) perform in mapping the inputs in Question 2.1 to the outputs in Question 2.2?

9. Data-to-text generation
13. Paraphrasing/lossless simplification
16. Summarisation (text-to-text)

2.4 What are the language(s) of the inputs accepted by the system(s)?

29. Chinese
41. English

2.5 What are the language(s) of the outputs produced by the system?

29. Chinese
41. English

3 Sample of System Outputs, Evaluators, Experimental Design

3.1 Sample of system outputs (test set)

Questions 3.1.1–3.1.3 record information about the size of the sample of outputs (or human-authored stand-ins) evaluated per system, how the sample was selected, and what its statistical power is.

3.1.1 How many system outputs (or other evaluation items) are evaluated per system in the evaluation experiment?

150.

3.1.2 How are system outputs (or other evaluation items) selected for inclusion in the evaluation experiment?

1. By an automatic random process.

3.1.3 Statistical power of the sample size.

3.1.3.1 What method was used to determine the statistical power of the sample size?

N/A.

3.1.3.2 What is the statistical power of the sample size?

N/A

3.1.3.3 Where can other researchers find details of the script used?

N/A

3.2 Evaluators

3.2.1 How many evaluators are there in this experiment?

3 evaluators.

3.2.2 Evaluator Type

3.2.2.1 Are the evaluators in this experiment domain experts?

2. No

3.2.2.2 Did participants receive any form of payment?

1. Paid (monetary compensation)

3.2.2.3 Were any of the participants previously known to the authors?

2. No

3.2.2.4 Were any of the researchers running the experiment among the participants?

2. No

3.2.3 How are evaluators recruited?

We recruited evaluators through the university's internal social media channels. Interested candidates were screened based on their academic qualifications and language proficiency certificates.

3.2.4 What training and/or practice are evaluators given before starting on the evaluation itself?

Before the formal evaluation, each annotator received a detailed instruction document that included:

1. The overall goal of the evaluation (assessing hallucination in generated text).
2. Clear definitions of hallucination, with examples specific to each of the three NLG tasks.
3. A walkthrough of the evaluation interface and the annotation protocol.
4. Five practice examples (with known correct labels). Annotators completed these practice items independently and then discussed their answers with the authors to resolve any misunderstandings

3.2.5 What other characteristics do the evaluators have?

All three evaluators are native Mandarin speakers but have near-native English proficiency. They are full-time master's students in English Language and Literature or Translation Studies at a Chinese university. All have passed the Test for English Majors Grade Eight with "Excellent". Their average age is 24; two are female, one is male.

3.3 Experimental Design

3.3.1 Has the experimental design been preregistered?

2. No

3.3.2 By what medium are responses collected?

Local spreadsheet files.

3.3.3 Quality assurance

Notes: Question 3.3.3.1 records information about the type(s) of quality assurance employed, and Question 3.3.3.2 records the details of the corresponding quality assurance methods.

3.3.3.1 What types of quality assurance methods are used to ensure that evaluators are sufficiently qualified and/or that their responses are of sufficient quality?

2. Automatic quality checking methods are used during and/or after evaluation.

6. Other: Structured pre-evaluation training with practice examples and explicit feedback, combined with a detailed annotation guideline document.

3.3.3.2 What methods are used for each of the types of quality assurance methods that were selected in Question 3.3.3.1?

For manual checking, the first author reviewed each annotator's response patterns for anomalies and spot-checked 10 random instances per task to verify label consistency. For the "Other" method, all evaluators completed mandatory training with detailed guidelines, 15 practice examples, and a group discussion to calibrate understanding before the main evaluation.

3.3.4 Form/Interface

3.3.4.1 Where can the form/interface that was shown to participants be viewed?

N/A

3.3.4.2 What types of information are evaluators shown when carrying out evaluations?

The task instruction, the reference output, and the corresponding system output generated by Qwen2.5-72B-Instruct.

3.3.5 How free are evaluators regarding when and how quickly to carry out evaluations?

2. Evaluators must complete each individual assessment within a set amount of time.
4. Evaluators must complete the whole evaluation in one sitting.

3.3.6 Are evaluators told they can ask questions about the evaluation and/or provide feedback?

3. Evaluators provide feedback after the evaluation.

3.3.7 What are the conditions in which evaluators carry out the evaluations?

1. Evaluators carry out assessments at a place of their own choosing.

3.3.8 In what ways do conditions in which evaluators carry out the evaluations vary for different evaluators?

N/A

4 Definition and Operationalisation of Quality Criteria

Questions in this section record information about each quality criterion (Fluency, Grammaticality, etc.) assessed in the human evaluation experiment that this sheet is being completed for.

4.1 Quality Criterion Properties

Questions 4.1.1-4.1.3 capture aspects of quality assessed by a given quality criterion in terms of three orthogonal properties: (i) what type of quality is being assessed; (ii) what aspect of the system output is being assessed; and (iii) whether system outputs are assessed in their own right or with reference to some system-internal or system-external frame of reference. For full explanations see ?.

4.1.1 What type of quality is assessed by the quality criterion?

1. Correctness.

4.1.2 Which aspect of system outputs is assessed by the quality criterion?

2. Content of output.

4.1.3 Is each output assessed for quality in its own right, or with reference to a system-internal or external frame of reference?

2. Quality of output relative to the input.

4.2 Evaluation Mode Properties

4.2.1 Does an individual assessment involve an objective or a subjective judgment?

2. Subjective.

4.2.2 Are outputs assessed in absolute or relative terms?

1. Absolute.

4.2.3 Is the evaluation intrinsic or extrinsic?

1. Intrinsic.

4.3 Response Elicitation

The questions in this section concern response elicitation, by which we mean how the ratings or other measurements that represent assessments for the quality criterion in question are obtained. This includes what is presented to evaluators, how they select a response, and via what type of tool, etc.

4.3.1 Quality criterion name.

4.3.1.1 What do you call the quality criterion in explanations/interfaces to evaluators?

Hallucination.

4.3.1.2 What standardised quality criterion name does the name entered for 4.3.1.1 correspond to?

Factual Consistency.

4.3.2 What definition do you give for the quality criterion in explanations/interfaces to evaluators?

Hallucination: The generation of content that appears plausible and coherent but lacks grounding in the provided source input or established factual knowledge.

4.3.3 What is the size of the scale or other rating instrument?

- 2.

4.3.4 What are the possible values of the scale or other rating instrument?

Two mutually exclusive binary categories: "Yes" and "No " .

4.3.5 How is the scale or other rating instrument presented to evaluators?

1. Multiple-choice options.

4.3.6 If there is no rating instrument, what is the task the evaluators perform?

N/A

4.3.7 What is the verbatim question, prompt or instruction given to evaluators (visible to them during each individual assessment)?

"For each row below, you will find the task instruction, the source input (original text / structured data/article), and the corresponding system output generated by the model. Please read each system output carefully and compare it against the task instruction and the reference output. Based on the definition of hallucination provided in your training materials, decide whether the system output contains any hallucinated content. Type 'Yes' if it does, or 'No' if it does not, in the empty column labelled 'Judgment' for each row. Please complete all 50 rows."

4.3.8 What form of response elicitation is used in collecting assessments from evaluators?

7. Output classification or labelling.

4.3.9 How are raw responses from participants aggregated or otherwise processed to obtain reported scores for this quality criterion?

For each task's 50 evaluation instances, we applied a majority-vote rule: a model output was deemed hallucinated when at least two of the three raters flagged it as such.

4.3.10 What method(s) are used for determining effect size and significance of findings for this quality criterion?

None.

4.3.11 Inter-annotator agreement.

4.3.11.1 How was the inter-annotator agreement between evaluators

measured for this quality criterion?

Fleiss' Kappa.

4.3.11.2 What was the inter-annotator agreement score?

Paraphrasing: 0.7580

Summarisation: 0.7770

Data-to-text: 0.7083

4.3.12 Intra-annotator agreement.

4.3.12.1 How was the intra-annotator agreement between evaluators measured for this quality criterion?

IntraAA not assessed

4.3.12.2 What was the intra-annotator agreement score?

IntraAA not assessed

5 Ethics

Questions in this section record information about the ethics approval, personal data, special category data, and impact assessments relating to the evaluation experiment.

5.1 Which research ethics committee has approved the evaluation experiment this sheet is being completed for, or the larger study it is part of?

No ethical approval obtained

5.2 Does personal data (as defined in GDPR Art.4, §1: <https://gdpr.eu/article-4-definitions>) occur in any of the system outputs (or human-authored stand-ins) evaluated, or responses collected, in the experiment this sheet is being completed for?

1. No, personal data as defined by GDPR was neither evaluated nor collected.

5.3 Does special category information (as defined in GDPR Art.9, §1: <https://gdpr.eu/article-9-processing-special-categories-of-personal-data-prohibited>) occur in any of the evaluation items evaluated, or responses collected, in the evaluation experiment this sheet is being completed for?

1. No, special category data as defined by GDPR was neither evaluated nor collected.

5.4 Have any impact assessments been carried out for the evaluation experiment, and/or any data collected/evaluated in connection with it?

No impact assessment carried out.